# Exercise Sheet 21

The notation is that of Lecture 21. The space $V = \mathbb{C}^{2n}$ has basis

$$e_1, \dots, e_n,\ f_1, \dots, f_n$$

and symplectic form $\omega$. The operators are

$$L = \Omega \wedge (\cdot), \qquad \Omega = \sum_{i=1}^{n} e_i \wedge f_i, \qquad \iota_\omega = \sum_{i=1}^{n} \iota_{f_i^*}\iota_{e_i^*}, \qquad H|_{\Lambda^k V} = n - k.$$

Exercises 21.1–21.3 develop fluency. Exercises 21.4 and 21.5 reconstruct and extend the proofs.

**Exercise 21.1 (The $C_3$ data).** Let $n = 3$.

(i) Write each of the nine positive roots in terms of $\alpha_1, \alpha_2, \alpha_3$, and check that half their sum is $\rho = 3\varepsilon_1 + 2\varepsilon_2 + \varepsilon_3$.

(ii) Write the Cartan matrix $(a_{ij}) = (\alpha_j(h_i))$.

(iii) Compute $\dim L(\omega_k)$ for $k = 1, 2, 3$ directly from the Weyl product, and compare with $\binom{6}{k} - \binom{6}{k-2}$.

**Exercise 21.2 (A basis-free contraction formula).**

(i) Prove that for $v_1, \dots, v_k \in V$,

$$\iota_\omega(v_1 \wedge \cdots \wedge v_k) = \sum_{p<q} (-1)^{p+q+1} \omega(v_p, v_q)\, v_1 \wedge \cdots \wedge \widehat{v_p} \wedge \cdots \wedge \widehat{v_q} \wedge \cdots \wedge v_k.$$

(ii) Use (i) to give a second proof that $\iota_\omega$ commutes with $\rho_\wedge(X)$ for $X \in \mathfrak{sp}_{2n}$.

**Exercise 21.3 (Explicit contraction matrices for $\mathfrak{sp}_4$).** Let $n = 2$. Use the bases

$$\Lambda^1 V:\ (e_1, e_2, f_1, f_2),$$
$$\Lambda^3 V:\ (b_1, b_2, b_3, b_4) = (e_1 \wedge e_2 \wedge f_1,\ e_1 \wedge e_2 \wedge f_2,\ e_1 \wedge f_1 \wedge f_2,\ e_2 \wedge f_1 \wedge f_2),$$

the basis of $\Lambda^2 V$ from Lecture 21, and $\mathrm{vol} = e_1 \wedge f_1 \wedge e_2 \wedge f_2$ for $\Lambda^4 V$.

(i) Find the matrices $D(L)$ of $L\colon \Lambda^1 V \to \Lambda^3 V$ and $D(\iota_\omega)$ of $\iota_\omega\colon \Lambda^3 V \to \Lambda^1 V$.

(ii) Find the matrices of $L$ and $\iota_\omega$ between $\Lambda^0 V$, $\Lambda^2 V$ and $\Lambda^4 V$.

(iii) Verify $[\iota_\omega, L] = 2 - k$ on each $\Lambda^k V$, and find $\Lambda^k_0 V$ for $k = 3, 4$.

**Exercise 21.4 (Primitive $\Lambda^3(\mathbb{C}^6)$).** Let $n = 3$.

(i) List the weights of $\Lambda^3 V$ with multiplicities.

(ii) Deduce the weights of $\Lambda^3_0 V$ and check that its dimension is 14.

(iii) Find a basis of the weight-$\varepsilon_1$ space of $\Lambda^3 V$, a nonzero primitive vector in it, and the vector $L(e_1) = \Omega \wedge e_1$.

(iv) Show that $X_{32} Y_{33}^{\mathsf{T}}(e_1 \wedge e_2 \wedge e_3)$ is that primitive vector up to a nonzero scalar, and state the decomposition of $\Lambda^3 V$.

**Exercise 21.5 (Duality and invariant forms).** Let $0 \le k \le n$. The vector

$$\Omega^n = n!\, e_1 \wedge f_1 \wedge \cdots \wedge e_n \wedge f_n$$

spans $\Lambda^{2n}V$.

(i) Prove that $L^{n-k}\colon \Lambda^k V \to \Lambda^{2n-k}V$ is an isomorphism of $\mathfrak{sp}_{2n}$-modules.

(ii) Define the bilinear form $B_k$ on $\Lambda^k V$ by

$$\Omega^{n-k} \wedge x \wedge y = B_k(x,y)\,\Omega^n.$$

Prove that $B_k$ is invariant, symmetric for $k$ even and skew for $k$ odd, and nondegenerate on $\Lambda^k V$.

(iii) Prove that $B_k$ is nondegenerate on $\Lambda^k_0 V$. Deduce $L(\omega_k)^* \cong L(\omega_k)$.

(iv) Check this against the formula $L(\lambda)^* \cong L(-w_0\lambda)$ of Lecture 14 by showing $w_0 = -1$ in type $C_n$.